

- 2) What are the main features of new-classical economics?

.....

.....

.....

.....

1.4 DETERMINATION OF OUTPUT AND EMPLOYMENT

The classical theory is based on the following assumptions:

- a) firms and workers are optimisers,
- b) they have perfect knowledge, and
- c) they operate in competitive (perfect) markets.

An implication of the above is that wages and prices are fully flexible. Under perfect competition, the demand for labour for the profit maximising firms is

$$P = W/MPN \quad \dots (1.1)$$

where, P is the product price,

W is nominal wages, and

MPN is the marginal product of labour.

As there is perfect competition, P is equal to Marginal Revenue (MR) which is received from the sale of one unit of output. Here, W/MPN represents the marginal cost of production, that is, the cost of production of an additional unit of output.

Equation (1.1) can be rewritten as

$$MPN = W/P \quad \dots (1.2)$$

(We denote nominal wage by W and real wage by its lower case, i.e., w)

Equation (1.2) indicates that the marginal product of labour is equal to the real wage (W/P). Hence, the labour demand curve in terms of real wage is nothing but the Marginal Product of Labour. The labour demand curve is downward sloping.

$$MPN_d = f(W/P) \quad \dots (1.3)$$

The supply of labour varies positively with the real wage. It assumes that individual labour maximises its utility.

This can be written as

$$N_s = g(W/P) \quad \dots (1.4)$$

where N_s indicates the supply of labour. We explain below in Fig. 1.1 the equilibrium in the labour market. Note that the labour market is in equilibrium at full employment of labour. At the equilibrium point, the supply of labour is equal to demand for labour. In the upper panel, we provide labour supply and labour demand as functions of real wage $\left(\frac{W}{P}\right)$. In the lower panel, we provide labour supply and labour demand as functions of nominal wage. As price level

increases from P_1 to P_2 and then to P_3 , real wage can be maintained at the same level if nominal wage increases to $2W$ and $3W$ respectively. The shifts in the supply curve of labour and demand curve of labour are shown in panel (b) of Fig. 1.1. Notice that there is no change in labour supply, as full employment is ensured (since there is no change in real wage). If there is an increase in output prices, but no increase in nominal wage rate, there will be a decline in real wage. This will lead to a decrease in labour supply.

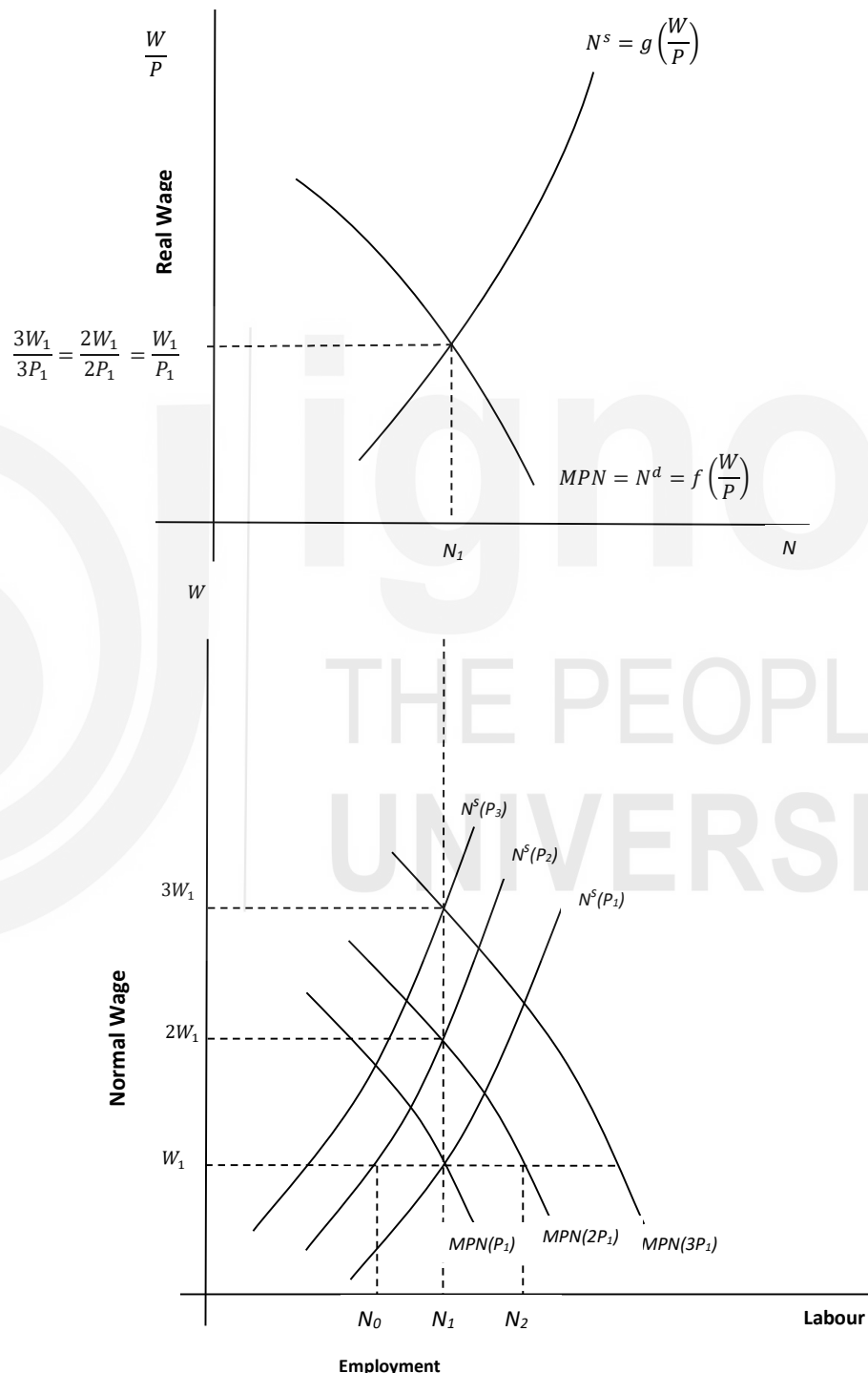


Fig.1.1: Supply of and Demand for Labour

Let us assume that there are only two factors of production, i.e., labour and capital. Output is determined by the production function $Y=F(K,N)$ as given in Fig. 1.2, panel (b). Equilibrium in the labour market (see panel (a) of Fig. 1.2) determines the full employment labour supply and real wage rate. Based on that full employment output is given in panel (b) of Fig. 1.2. As you can see, the economy operates with full employment output (Y_0) and full employment labour supply (N_0).

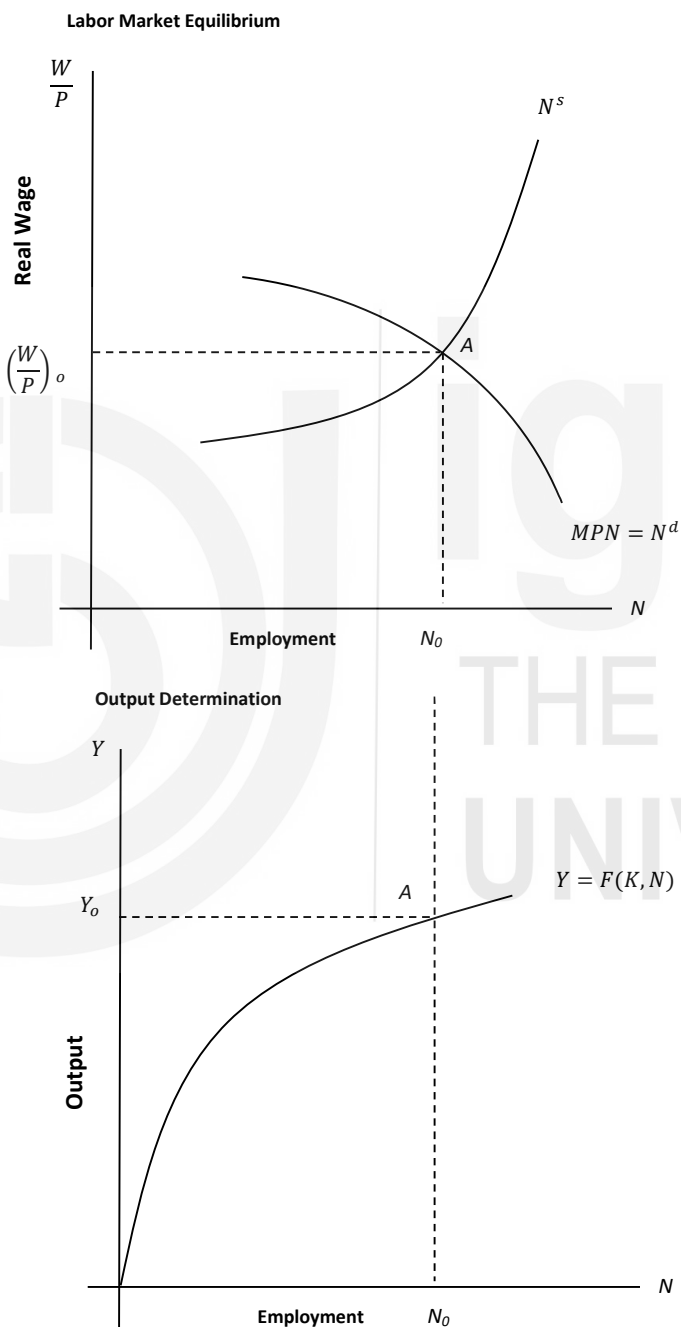


Fig. 1.2: Equilibrium Output and Employment